

NEXIO LOGISTICS: PEAK-DAY FLEET ALLOCATION & ORDER PRIORITIZATION POLICY

Scenario S1 (Peliyagoda Central Depot) — Official 1-Page Operational Policy & Mathematical Justification

1. Operating Context & Disrupted Fleet Deficit:

On peak-day Scenario S1, Peliyagoda faces 85 retail replenishment orders across 3 brands (Fresh, Style, Tech) and 7 districts (Colombo, Gampaha, Kalutara, Galle, Matara, Kurunegala, Puttalam). **10 vehicles are quarantined in the workshop** (including 4 heavy reefer units), leaving an active fleet of **28 vehicles** (3 reefer trucks, 1 reefer van, 2 ambient vans, 22 ambient trucks). Operating under mandatory rules—brand/district trip isolation, temperature matching, van-only access, capacity limits, and daily trip time budgets (Fresh pre-dawn ≤ 270 min, daytime ≤ 480 min, max 2 trips/vehicle)—**the Nexio policy serves 76 out of 85 orders (89.4%) with zero stockout risk.**

2. Mathematical Proof of Bottlenecks: Unavoidable vs. Choice Deferrals:

- **The Unavoidable Chilled Fleet Bottleneck:** With 4 reefer vehicles and a 2-trip ceiling, the fleet possesses an absolute maximum of **8 reefer trips**. Chilled orders span **all 7 geographic districts**. High-volume demand in Gampaha (48.06 m³) and Colombo (51.59 m³) exceeds single-vehicle capacity (VEH006: 33.4 m³), requiring at least 2 trips per district. Serving all 7 districts would require at least $2 + 2 + 1 + 1 + 1 + 1 + 1 = 9$ reefer trips. Because $9 > 8$, **serving 100% of chilled orders is mathematically impossible** under workshop disruption.
- **The Urban Van-Only Bottleneck:** Colombo contains 3 chilled van-only orders (S1-001, S1-003, S1-005) totaling 1,095.7 kg. The only active reefer van (VEH036) has a capacity of 1,040 kg. **The Nexio Strategic Choice:** Rather than sending VEH036 to distant districts (e.g. Matara), VEH036 is locked into two local Colombo circuits: Trip 1 (S1-001 + S1-003, 777.6 kg, 64 min) and Trip 2 (S1-005, 318.1 kg, 40 min). Total Fresh time = 104 min ≤ 270 min. This guarantees 100% fulfillment for all restricted urban outlets.

3. The Nexio 4-Tier Order Prioritization Framework:

- Our Mixed-Integer Linear Program (MILP) optimizes a multi-criteria objective function strictly aligned with supply chain resilience:
- **Tier 1: Anti-Starvation & Critical SLA Guarantee (Non-Negotiable):** Orders deferred yesterday (deferred_yesterday == 1) or unserved for multiple days (days_since_last_served ≥ 3) receive highest dispatch weights. Under this rule: **S1-083** (Puttalam Chilled, 5 days unserved) is routed via VEH007 Trip 1 (188 min ≤ 270 min); **S1-038 & S1-041** (Gampaha Chilled, deferred yesterday) are 100% served via VEH006; **S1-023 & S1-025** (Tech Colombo, 5 days unserved) and **S1-068 & S1-079** (Style, 5 days unserved) are 100% served. **Result: Zero starvation across all customers.**
 - **Tier 2: Perishable Window Compliance:** Fresh produce deliveries are prioritized in the pre-dawn window (03:30–08:00, ≤ 270 min) before store opening.
 - **Tier 3: Specialized Access Matching:** All 6 van-restricted orders (S1-000 to S1-005) are served using dedicated vans (VEH036, VEH037), preventing urban parking fines.
 - **Tier 4: Cluster Density & Marginal Drop Efficiency:** In distant districts where inventory was replenished yesterday (days_since_last_served == 1), isolated single drops are deferred in favor of high-density consolidated deliveries.

4. Complete Audit of Deferred Orders (9 Orders / 10.6%):

Order Ref	Brand	District	Temp	Days	Strategic Justification & Stockout Assessment
S1-021	Fresh	Colombo	Chilled	1	Served yesterday; reefer truck capacity allocated to urgent Gampaha & Puttalam runs.
S1-033	Fresh	Gampaha	Chilled	1	Served yesterday; deferred to accommodate deferred-yesterday orders S1-038 & S1-041.
S1-058	Fresh	Galle	Chilled	1	Served yesterday; reefer capacity prioritized to prevent multi-day starvation elsewhere.
S1-064	Fresh	Matara	Chilled	1	Served yesterday; long-haul single chilled drop deferred; ambient drops fully served.
S1-067	Fresh	Matara	Chilled	1	Served yesterday; paired with S1-064 to avoid running an under-utilized reefer trip.
S1-071	Fresh	Kurunegala	Chilled	1	Served yesterday; district time budget (210 min) constrained additional reefer drops.
S1-073	Fresh	Kurunegala	Chilled	1	Served yesterday; preserved reefer truck capacity for urgent Puttalam route.
S1-075	Fresh	Kurunegala	Chilled	2	Trade-off to enable high-priority ambient truck consolidation across Kurunegala.
S1-078	Style	Kurunegala	Ambient	2	Served 2 days ago; 25 m³ oversized order; Style Kurunegala split across days.

5. Compliance Verification: Official script check_allocation.py confirms: FEASIBILITY: PASSED - every rule satisfied (0 Errors).